

If you can't read this you need to get closer



What are these?

Sea Anemones

Coral and Sea

anemone

Cnidaria
Anthozoa



What are they?

Anthozoans are types of sea **animals** that have lived on earth for over 500 million years, They belong to the Phylum Cnidarians with jellyfish
They are simple life forms and not every complex, they spend their whole lives as Polyps



How do they live

Life



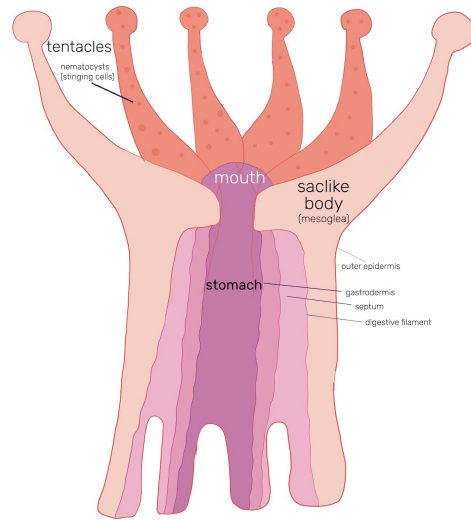
Food

Coral - They get food from the algae living inside them. They also eat plankton in the water

Sea anemones - They are larger than coral so they eat bigger prey like small fish and they also eat plankton



What are Coral really?

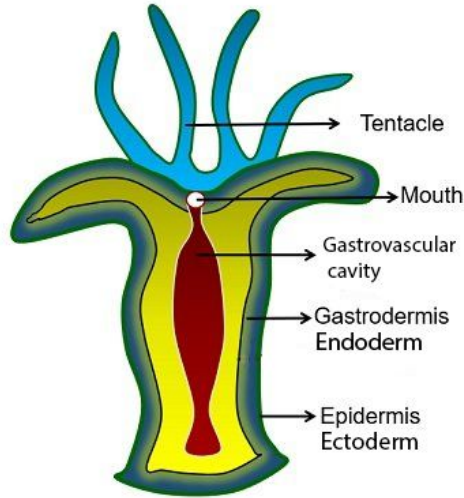


Each of these is in reality a single organism that has grown a body with a layer of animals that form a colony specialized for coral as well

←
Coral and Sea anemones take the adult form of a polyp, opposite of jellyfishes



Polyp Anatomy



Tentacles: used to grab food from the water

Mouth: Entrance to its “stomach”, used to swallow food

In truth they are also transparent, not colorful like they look. Then why are they so colorful, it's because of algae that live inside the coral.

Gastrodermis: Endoderm - inner layer, lines the Gastrovascular cavity, the one of the two body's wall

Epidermis: Ectoderm - outer layer, skin





Coral and Algae



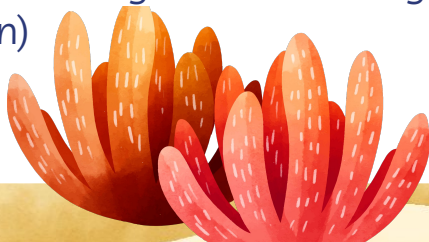
But what happens if the landlord can't afford to continue housing their renter? They kick them out.

- Nutrients from algae photosynthesis
- Oxygen from the algae
- Waste management from algae (Carbon)



Qualistic benefit
The coral provides its renter with a variety in colors

- Shelter from predators in the coral
Think of it as a renter and landlord, the coral provides the home and the algae provides food.



Coral bleaching and what causes it



This is called coral bleaching. When coral kicks out the algae that gives it its color and it loses its main source of food, it becomes more vulnerable to diseases and can die.

This is the first response coral has to stress and it happens in response to something, in this case, changes to the water.



Global warming and its problems for coral



So far estimates say global warming has risen temperatures in the ocean by 1.5 F degrees.

if temperatures rise by 2.7 F up to 70-90% of coral reefs would be at risk of death

Global warming is a huge threat to coral reefs, 50% of them have died in 70 years and more will follow.



Why should you care?



They also absorb waves, reducing erosion and protecting coastlines from storms, they give amazing views and help provide many thousands of species. Taking that away would put many species at risk of extinction.

The coral reefs house around 25% of marine life, or hundreds of thousands of species. Taking that away would put many species at risk of extinction.

They also absorb waves, reducing erosion and protecting coastlines from storms, they give amazing views and help provide many thousands of species. Taking that away would put many species at risk of extinction.

They have also been used for medicine for diseases like asthma and cancer. And if we don't do something about it we could lose these beautiful structures that help provide so much for us

